

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\ln(\tan x)}{\tan x} - \frac{\ln(\sec x)}{\tan x} dx$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\ln(\frac{\sin x}{\cos x})-\ln(\frac{1}{\cos x})}{\tan x} dx$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\ln\left(\frac{\sin x}{\frac{1}{\cos x}}\right)}{\tan x} dx$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos x \ln(\sin x)}{\sin x} dx$$

$$u:=\sin x$$

$$du=\cos x dx$$

$$\int_{\frac{1}{\sqrt{2}}}^1 \frac{\ln u}{u} du$$

$$t:=\ln u$$

$$dt=\frac{1}{u} du$$

$$\int_{-\frac{\ln 2}{2}}^0 t dt$$

$$\begin{aligned} &\frac{t^2}{2} \Big|_{-\frac{\ln 2}{2}}^0 \\ &-\frac{\left(-\frac{\ln 2}{2}\right)^2}{2} \\ &-\frac{\ln^2 2}{8} \end{aligned}$$